


$$\frac{969}{57} = ?$$

$$\frac{1058}{23} = ?$$

$$\frac{3526}{41} = ?$$

$$\frac{1335}{89} = ?$$

$$\frac{3182}{37} = ?$$

$$\frac{1032}{43} = ?$$

$$\begin{array}{r}
 17 \\
 \Rightarrow \begin{array}{r}
 \cancel{323} \\
 \hline
 969 \\
 \hline
 57 \\
 \hline
 19
 \end{array}
 \end{array}$$

17 ✓

$$\begin{array}{r}
 \textcircled{15} \\
 \hline
 \cancel{1335} \\
 \hline
 89
 \end{array}$$

$$\begin{array}{r}
 46 \\
 \hline
 \textcircled{1058} \\
 \hline
 23
 \end{array}$$

unit
digit

$$\begin{array}{r}
 \boxed{86} \\
 \hline
 3182 \\
 \hline
 37
 \end{array}$$

$$\begin{array}{r}
 86 \\
 \hline
 \cancel{3826} \\
 \hline
 \textcircled{41}
 \end{array}$$

$$\begin{array}{r}
 \boxed{24} \\
 \hline
 \textcircled{1032} \\
 \hline
 4\boxed{3}
 \end{array}$$

$$\begin{array}{r}
 \textcircled{155} \\
 \hline
 \boxed{2635} \\
 \hline
 17
 \end{array}$$

$$\frac{4606}{47} = ?$$

✓✓

$$\frac{1044}{29} = ?$$

$$\frac{3498}{53} = ?$$

$$\frac{3886}{67} = ?$$

$$\begin{array}{r}
 98 \\
 \boxed{46} \overline{) 466} \\
 \underline{47} \\
 =
 \end{array}$$

$$\begin{array}{r}
 \boxed{66} \\
 \underline{3498} \\
 53
 \end{array}$$

$$\begin{array}{r}
 36 \\
 \boxed{104} \overline{) 4} \\
 \underline{29}
 \end{array}$$

$$\begin{array}{r}
 \textcircled{58} \\
 \underline{(388)6} \\
 67
 \end{array}$$

Square Numbers from 1-100

$1^2 = 1$	$21^2 = 441$	$41^2 = 1681$	$61^2 = 3721$	$81^2 = 6561$
$2^2 = 4$	$22^2 = 484$	$42^2 = 1764$	$62^2 = 3844$	$82^2 = 6724$
$3^2 = 9$	$23^2 = 529$	$43^2 = 1849$	$63^2 = 3969$	$83^2 = 6889$
$4^2 = 16$	$24^2 = 576$	$44^2 = 1936$	$64^2 = 4096$	$84^2 = 7056$
$5^2 = 25$	$25^2 = 625$	$45^2 = 2025$	$65^2 = 4225$	$85^2 = 7225$
$6^2 = 36$	$26^2 = 676$	$46^2 = 2116$	$66^2 = 4356$	$86^2 = 7396$
$7^2 = 49$	$27^2 = 729$	$47^2 = 2209$	$67^2 = 4489$	$87^2 = 7569$
$8^2 = 64$	$28^2 = 784$	$48^2 = 2304$	$68^2 = 4624$	$88^2 = 7744$
$9^2 = 81$	$29^2 = 841$	$49^2 = 2401$	$69^2 = 4761$	$89^2 = 7921$
$10^2 = 100$	$30^2 = 900$	$50^2 = 2500$	$70^2 = 4900$	$90^2 = 8100$
$11^2 = 121$	$31^2 = 961$	$51^2 = 2601$	$71^2 = 5041$	$91^2 = 8281$
$12^2 = 144$	$32^2 = 1024$	$52^2 = 2704$	$72^2 = 5184$	$92^2 = 8464$
$13^2 = 169$	$33^2 = 1089$	$53^2 = 2809$	$73^2 = 5329$	$93^2 = 8649$
$14^2 = 196$	$34^2 = 1156$	$54^2 = 2916$	$74^2 = 5476$	$94^2 = 8839$
$15^2 = 225$	$35^2 = 1225$	$55^2 = 3025$	$75^2 = 5625$	$95^2 = 9025$
$16^2 = 256$	$36^2 = 1296$	$56^2 = 3136$	$76^2 = 5776$	$96^2 = 9216$
$17^2 = 289$	$37^2 = 1369$	$57^2 = 3249$	$77^2 = 5929$	$97^2 = 9409$
$18^2 = 324$	$38^2 = 1444$	$58^2 = 3364$	$78^2 = 6084$	$98^2 = 9604$
$19^2 = 361$	$39^2 = 1521$	$59^2 = 3481$	$79^2 = 6241$	$99^2 = 9801$
$20^2 = 400$	$40^2 = 1600$	$60^2 = 3600$	$80^2 = 6400$	$100^2 = 10000$

Identifying Square Roots of Perfect Numbers

46 47 48 49

$$1 \rightarrow 1$$

$$2 \rightarrow 4$$

$$3 \rightarrow 9$$

$$4 \rightarrow 16$$

$$5 \rightarrow 25$$

$$6 \rightarrow 36$$

$$7 \rightarrow 49$$

$$8 \rightarrow 64$$

$$9 \rightarrow 81$$

$$45^2 - 50^2 \leftarrow \sqrt{2116} = 46$$

$$70^2 - 75^2 \leftarrow \sqrt{5329} = 73$$

$$65^2 - 70^2 \leftarrow \sqrt{4489} = 67$$

$$85^2 - 90^2 \leftarrow \sqrt{7396} = 86$$

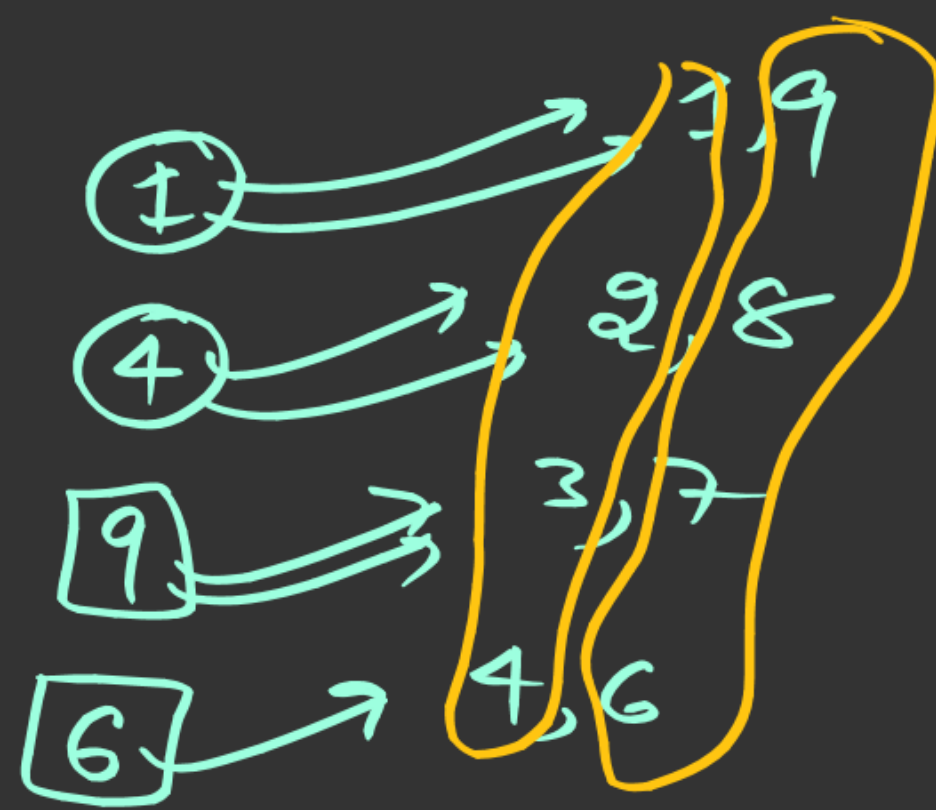
$$90^2 - 95^2 \leftarrow \sqrt{8464} = 92$$

$$55^2 - 60^2 \leftarrow \sqrt{3481} = 57$$

$$50^2 - 55^2 \leftarrow \sqrt{2916} = 54$$

$$75^2 - 80^2 \leftarrow \sqrt{6084} = 78$$

$$65^2 - 70^2 \leftarrow \sqrt{4761} = 69$$



*

Identifying Square Roots of Non-Perfect Numbers

#

$$\sqrt{a \pm b} = \underbrace{(\sqrt{a})}_{\text{Perfect Square Part}} \pm \underbrace{\frac{b}{2\sqrt{a}}}_{\text{extra part}}$$

Perfect
Square
Part

extra
part

$$36.25$$

$$\sqrt{210} = \sqrt{196 + 14} = 14 + \frac{14}{28} \Rightarrow 14.5$$

$$\sqrt{264} = \sqrt{256 + 8} = 16 + \frac{8}{32} \Rightarrow 16.25$$

$$\sqrt{568} = \sqrt{576 - 8} = 24 - \frac{8}{48} \Rightarrow 24 - 0.166 = 23.8$$

$$\sqrt{717} = \sqrt{729 - 12} = 27 - \frac{12}{549} \Rightarrow 27 - 0.2222 = 26.777$$

$$\sqrt{1314} = \sqrt{1296 + 18} = 36 + \frac{18}{72} \Rightarrow 36.25$$

Practice sheets ✓✓

- ① ✓
- ② ✓
- ③ ✓

Cubes Unit digit

$$\sqrt[3]{2198}$$

$$\sqrt[3]{2197} \Rightarrow 13$$

Cube Numbers from 1 to 25

$1^3 = 1$	$6^3 = 216$	$11^3 = 1331$	$16^3 = 4096$	$21^3 = 9261$
$2^3 = 8$	$7^3 = 343$	$12^3 = 1728$	$17^3 = 4913$	$22^3 = 10648$
$3^3 = 27$	$8^3 = 512$	$13^3 = 2197$	$18^3 = 5832$	$23^3 = 12167$
$4^3 = 64$	$9^3 = 729$	$14^3 = 2744$	$19^3 = 6859$	$24^3 = 13824$
$5^3 = 125$	$10^3 = 1000$	$15^3 = 3375$	$20^3 = 8000$	$25^3 = 15625$

$1^3 \rightarrow 1$
 $2^3 \rightarrow 8$
 $3^3 \rightarrow 27$
 $4^3 \rightarrow 64$
 $5^3 \rightarrow 125$
 $6^3 \rightarrow 216$
 $7^3 \rightarrow 343$
 $8^3 \rightarrow 512$
 $9^3 \rightarrow 729$
 $10^3 \rightarrow 1000$
 $11^3 \rightarrow 1331$
 $12^3 \rightarrow 1728$
 $13^3 \rightarrow 2197$
 $14^3 \rightarrow 2744$
 $15^3 \rightarrow 3375$
 $16^3 \rightarrow 4096$
 $17^3 \rightarrow 4913$
 $18^3 \rightarrow 5832$
 $19^3 \rightarrow 6859$
 $20^3 \rightarrow 8000$
 $21^3 \rightarrow 9261$
 $22^3 \rightarrow 10648$
 $23^3 \rightarrow 12167$
 $24^3 \rightarrow 13824$
 $25^3 \rightarrow 15625$

2	8
3	7



$$\begin{aligned} 1^3 &\rightarrow 1 \\ 4^3 &\rightarrow 4 \\ 5^3 &\rightarrow 5 \\ 6^3 &\rightarrow 6 \\ 9^3 &\rightarrow 9 \end{aligned}$$

Identifying Cube Roots of Perfect Numbers

25 ↑

$$\sqrt[3]{5832} =$$

1	8
---	---

$$\sqrt[3]{17576} =$$

2	6
---	---

$$\sqrt[3]{54872} =$$

3	2
---	---

$$\sqrt[3]{175616} =$$

5	6
---	---

$$\sqrt[3]{32768} =$$

32

$$\sqrt[3]{250047} =$$

6	3
---	---

$$\sqrt[3]{658503} =$$

8	7
---	---

$$\sqrt[3]{205379} =$$

5	9
---	---

$$\sqrt[3]{804357} =$$

9	3
---	---

$$\sqrt[3]{456533} =$$

7	7
---	---

② $\Rightarrow$

Unit digit
0, 2, 4, 6, 8

③ $\Rightarrow$

Number



Sum of Digits



Obtained no. ✓

~~2~~

2 ✓

④ $\Rightarrow$

Last two
digits

$\Rightarrow$ "00"

or
divisible by 4

7(2)

82(2)

37(2)

82(4)

104(6)

96(6)

24(4)

12(8)

717(8)

912(6)

336(6)

42(4)

37(4)

2, 4, 6, 8, 10



123

$1+2+3 \Rightarrow 6$



712

$7+1+2 \Rightarrow 10 \times$



926

$9+2+6 \Rightarrow 17 \times$



114

$1+1+4 = 6$



837

$8+3+7 \Rightarrow 18$



11121

$1+1+1+2+1 = 6$

④

757216 ✓

212 ✓

12300 ✓

18926 ✗

12477 ✗

19896 ✓

214001 ✗

31249 ✓

81888 ✓

71392 ✓

102414

Last Two digits

Either

"00"

or

Divisible by 4

abc

$$100a + 10b + c$$

$$99a + a + 9b + b + c$$

$$\underbrace{(99a + 9b)}_{3 \checkmark} + \underbrace{(a + b + c)}_{3 ?}$$

abcd

$$1000a + 100b + 10c + d$$

$$\underline{999a} + a + \underline{99b} + b + \underline{9c} + c + d$$

$$\underbrace{(999a + 99b + 9c)}_{3 \checkmark} + \underbrace{(a + b + c + d)}_{3 ?}$$

757216

$$\boxed{757200} + \textcircled{16}$$

✓
↑ ✓

18926

$$\boxed{18900 + 26}$$

✓
↑ 4 ✓
9 9

$\boxed{5} \Rightarrow$
 unit digit
 $\downarrow$
 $\boxed{5,0}$

1235

3350

1220

17775

93790

$\boxed{6} = 2 \times 3$
 $\downarrow \quad \downarrow$
 $\checkmark \quad \checkmark$

2166

$\checkmark \quad \checkmark \quad \boxed{6} \checkmark$
 2 3

3362

$\checkmark \quad \times \quad \times$
 2 3 6

1298

$\checkmark \quad \times \quad \times$
 2 3 6

7, 13

8 $\Rightarrow$ ✓

Last three digits

"000"

or

"divisible by 8"

9 $\Rightarrow$

Sum of digits



Number

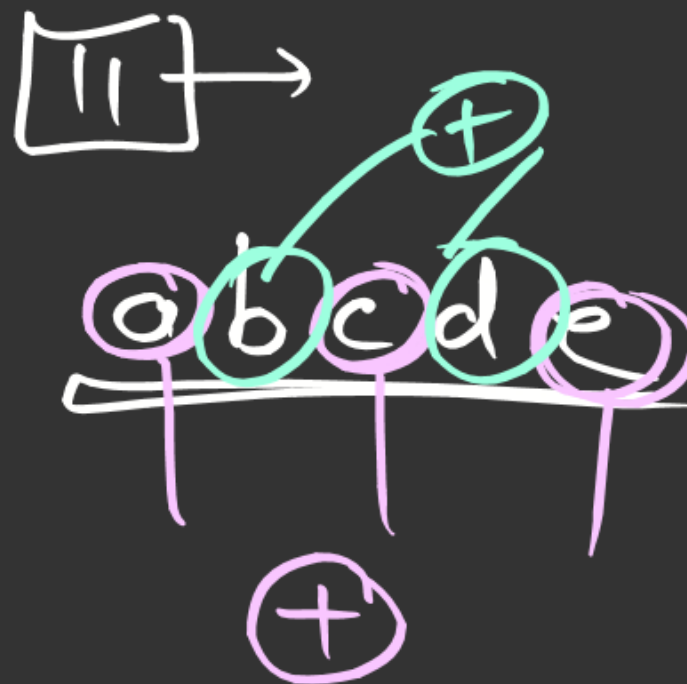


divisible by 9 ✓

10 ✓✓

Unit digit = 0

✓✓



0, 11, - - -

Divisibility Rules in case of 2, 4, 8 and 16

$$76 \div 2 =$$

$$156 \div 2 =$$

$$348 \div 2 =$$

$$7124 \div 2 =$$

$$236 \div 4 =$$

$$472 \div 4 =$$

$$692 \div 4 =$$

$$3156 \div 4 =$$

$$11244 \div 4 =$$

$$192 \div 8 =$$

$$368 \div 8 =$$

$$2776 \div 8 =$$

$$5928 \div 8 =$$

$$12944 \div 8 =$$

$$384 \div 16 =$$

$$896 \div 16 =$$

$$2464 \div 16 =$$

$$4032 \div 16 =$$

$$13408 \div 16 =$$

Divisibility Rules in case of 3 and 9

$$228 \div 3 =$$

$$555 \div 3 =$$

$$1782 \div 3 =$$

$$4278 \div 3 =$$

$$39756 \div 3 =$$

$$477 \div 9 =$$

$$891 \div 9 =$$

$$6885 \div 9 =$$

$$8748 \div 9 =$$

$$13788 \div 9 =$$

Divisibility Rules in case of 6, 12 and 15

$$774 \div 6 =$$

$$1068 \div 6 =$$

$$2142 \div 6 =$$

$$5148 \div 6 =$$

$$27834 \div 6 =$$

$$3852 \div 12 =$$

$$5904 \div 12 =$$

$$8148 \div 12 =$$

$$10068 \div 12 =$$

$$11724 \div 12 =$$

$$1065 \div 15 =$$

$$1935 \div 15 =$$

$$4185 \div 15 =$$

$$6555 \div 15 =$$

$$11145 \div 15 =$$

Divisibility Rules in case of 11

$$\begin{array}{r} 10 \\ \hline 10 \end{array}$$

$$\begin{array}{r} 6842 \\ \hline \end{array}$$

$$\boxed{17-6}$$

$$\boxed{11}$$

$$\begin{array}{r} 7,7 \\ \hline \end{array}$$

$$\begin{array}{r} 10 \\ \hline \end{array}$$

$$10-10$$

$$\Rightarrow 0$$

$$\begin{array}{r} 622 \\ \hline 6842 \div 11 = \end{array}$$

$$\begin{array}{r} 8294 \div 11 = \end{array}$$

$$12452 \div 11 =$$

$$19316 \div 11 =$$

$$31779 \div 11 =$$

$$182919 \div 11 =$$

$$\begin{array}{r} 26 \\ -4 \\ \hline 22 \\ \hline \end{array}$$

$$16629$$

Divisibility Rules in Specific Cases

$$\checkmark \underline{984} \leftarrow \frac{492}{\frac{1}{2}}$$

$$492 \div 0.5 =$$

$$\underline{456} \leftarrow \frac{228}{\frac{1}{2}}$$

$$684 \div 1.5 =$$

$$\textcircled{314} \leftarrow \frac{3140}{10}$$

$$\frac{785 \times 4}{10}$$

$$785 \div 2.5 =$$

$$628$$

$$1320 \div 1.25 =$$

$$\frac{2+3}{4} = \frac{5}{4}$$

$$\frac{3+1}{4} = \frac{4}{4}$$

$$935 \div 2.75 =$$

$$208 \div 3.25 =$$

$$\frac{157}{785 \times 4}$$

$$\frac{935}{785 \times 4}$$

$$\Rightarrow 340 \checkmark$$

Divisible
by 7

✓ 47 6 ✓✓

$$\begin{array}{r} 47 \\ -12 \\ \hline 35 \end{array}$$

✓ ⇒ 35

36 4

$$\begin{array}{r} 36 \\ -8 \\ \hline 28 \end{array}$$

✓✓

40 7

$$\begin{array}{r} 40 \\ -14 \\ \hline 26 \end{array}$$

✗

~~67 2~~

$$\begin{array}{r} 67 \\ -4 \\ \hline 63 \end{array}$$

✓

✓ 60 44(5)

$$\begin{array}{r} 6044 \\ -10 \\ \hline 6034 \end{array}$$

$$\begin{array}{r} 59 \\ -10 \\ \hline 49 \end{array} \checkmark$$

$$\begin{array}{r} 603 \\ -2 \\ \hline 598 \end{array}$$

4776(8)

$$\begin{array}{r} 4776 \\ -16 \\ \hline 4760 \end{array}$$

47(8)

$$\begin{array}{r} 47 \\ -12 \\ \hline 35 \end{array} \checkmark$$

⑫ — 3×4

$\frac{+ + +}{3}$ ✓

$\frac{0 0}{4}$, div. by 4

⑬

13 *

✓✓ 234

$$\begin{array}{r} 23 \\ + 16 \\ \hline 39 \end{array} \checkmark$$

✓✓ 4797

$$\begin{array}{r} 479 \\ + 28 \\ \hline 507 \end{array}$$

$$\begin{array}{r} 50 \\ 28 \\ \hline 78 \end{array} \checkmark$$

✗ 1106

$$\begin{array}{r} 110 \\ 24 \\ \hline 134 \end{array}$$

$$\begin{array}{r} 16 \\ 290 \end{array}$$

✓✓ 5928

$$\begin{array}{r} 592 \\ 32 \\ \hline 624 \end{array}$$

$$\begin{array}{r} 62 \\ 16 \\ \hline 78 \end{array} \checkmark$$

$$(14) \Rightarrow \boxed{2 \times 7}$$